

OE-EC804B — Microwave Integrated Circuits

Semester VIII • 3L:0T:0P • 3 Credits

Module 5 Exam Notes: MIC Measurement, Testing & Applications

CGEC | B.Tech ECE (2023–27) | Biraj Sarkar

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1 MIC Measurement Systems, Fixtures and Probes

Testing unpackaged MIC/MMIC chips presents unique challenges because standard coaxial cables (SMA, 2.92 mm, 1.85 mm) cannot interface directly with sub-millimeter planar microstrip pads on semiconductor wafers.

Microwave Test Fixture & On-Wafer Probes

A **Microwave Test Fixture** holds packaged or unpackaged MIC substrates and provides low-reflection transitions from coaxial launchers to planar microstrip traces. **On-Wafer RF Probes** (Ground-Signal-Ground / GSG coplanar probes) make direct electrical contact with MMIC pads on the semiconductor wafer using precision micro-positioning manipulators, operating up to 110 GHz.

1.1 Wafer Probe Architecture (GSG Configuration)

Coplanar GSG probes maintain a continuous $50\ \Omega$ coplanar transmission line right up to the beryllium-copper or tungsten probe tips, completely eliminating wire bond series inductance ($L_s \approx 0$) during wafer-level characterization.

2 S-Parameter Measurement and Calibration

2.1 Vector Network Analyzer (VNA)

A **Vector Network Analyzer (VNA)** measures both the magnitude and phase of 2-port scattering parameters ($S_{11}, S_{21}, S_{12}, S_{22}$) by transmitting a frequency-swept RF signal and separating incident, reflected, and transmitted waves using internal directional couplers and heterodyne receivers.

2.2 VNA Calibration & Systematic De-Embedding

Uncalibrated VNA measurements include systematic errors caused by cable attenuation, connector reflections, directional coupler directivity limits, and receiver frequency response. Calibration establishes mathematical error models to de-embed the fixture and move the measurement reference plane to the device under test (DUT).

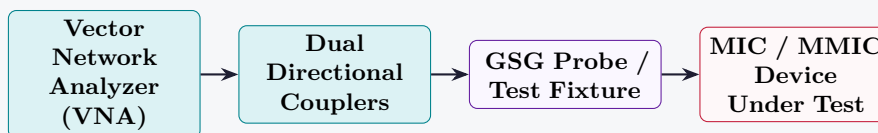
1. SOLT Calibration (Short-Open-Load-Thru):

- Standard calibration for coaxial measurements using 4 known standards: Short ($\Gamma = -1$), Open ($\Gamma = +1$), Matched Load ($\Gamma = 0$), and Thru line ($S_{21} = 1$).
- *Limitation in MMIC Testing:* Fabricating a precise, broadband $50\ \Omega$ matched load standard on a semiconductor substrate is extremely difficult due to parasitic capacitance.

2. TRL Calibration (Thru-Reflect-Line):

- Uses 3 simple planar standards printed directly on the substrate: a **Thru** line (0 length), an unknown high-reflection **Reflect** standard (Short or Open with identical $|\Gamma| \approx 1$), and a **Line** standard ($\lambda_g/4$ longer than Thru).
- **Primary calibration method for MMIC wafer probing** because it does not require a $50\ \Omega$ matched load standard.

Vector Network Analyzer (VNA) S-Parameter Measurement Setup



3 Noise Figure Measurement Techniques

Noise Figure (NF) measures the degradation of Signal-to-Noise Ratio (SNR) as a signal passes through a microwave circuit:

$$F = \frac{\text{SNR}_{in}}{\text{SNR}_{out}} \quad ; \quad NF(\text{dB}) = 10 \log_{10}(F) \quad (1)$$

3.1 The Y-Factor Measurement Method

The **Y-Factor Method** is the industry standard technique for measuring Noise Figure using a calibrated noise source toggled between two known noise temperatures: **HOT** (T_H) and **COLD** (T_C).

Y-Factor Derivation & Noise Figure Equations

Noise power measured at receiver output for HOT and COLD states:

$$N_{\text{HOT}} = k(T_H + T_e)B \cdot G \quad ; \quad N_{\text{COLD}} = k(T_C + T_e)B \cdot G \quad (2)$$

Taking the ratio yields the dimensionless Y -factor:

$$Y = \frac{N_{\text{HOT}}}{N_{\text{COLD}}} = \frac{T_H + T_e}{T_C + T_e} \Rightarrow T_e = \frac{T_H - YT_C}{Y - 1} \quad (3)$$

The noise factor F relative to reference temperature $T_0 = 290 \text{ K}$ (27°C):

$$F = 1 + \frac{T_e}{T_0} = \frac{\text{ENR}}{Y - 1} \quad \text{where ENR} = \frac{T_H - T_0}{T_0} \text{ (Excess Noise Ratio)} \quad (4)$$

Worked Numerical: Noise Figure Calculation via Y-Factor Method

Problem: A noise source with an Excess Noise Ratio $\text{ENR} = 15.0 \text{ dB}$ is connected to a microwave amplifier. The measured output noise power ratio between HOT and COLD states is $Y = 8.0 \text{ dB}$. Calculate:

- The linear ENR and linear Y -factor.
- The Noise Factor F and Noise Figure NF in dB.

Solution:

- Convert values from dB to linear ratios:

$$\text{ENR}_{\text{linear}} = 10^{15/10} = 10^{1.5} = 31.6228 \quad (5)$$

$$Y_{\text{linear}} = 10^{8/10} = 10^{0.8} = 6.30957 \quad (6)$$

- Calculate Noise Factor F :

$$\begin{aligned} F &= \frac{\text{ENR}_{\text{linear}}}{Y_{\text{linear}} - 1} = \frac{31.6228}{6.30957 - 1} \\ &= \frac{31.6228}{5.30957} = 5.9558 \end{aligned} \quad (7)$$

Convert F to Noise Figure NF in dB:

$$NF = 10 \log_{10}(5.9558) = 7.749 \text{ dB} \approx 7.75 \text{ dB} \quad (8)$$

Quick Revision — Module 5

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Key Concepts and Metrics at a Glance

Vector Network Analyzer (VNA)	Measures magnitude & phase of 2-port S-parameters ($S_{11}, S_{21}, S_{12}, S_{22}$).
TRL Calibration	Thru-Reflect-Line method; preferred for MMIC wafer probes as no matched load standard is required.
GSG Wafer Probes	Ground-Signal-Ground coplanar probes providing $50\ \Omega$ contact directly to MMIC pads.
Y-Factor Noise Measurement	Toggles noise source between HOT (T_H) and COLD (T_C) states ($Y = N_{HOT}/N_{COLD}$).
Excess Noise Ratio (ENR)	$ENR = (T_H - T_0)/T_0$; specifies calibrated output power of RF noise head.

One-Line Definitions

- **VNA:** Microwave instrument measuring complex reflection and transmission parameters of RF networks.
- **GSG Probe:** High-frequency coplanar probe contacting MMIC wafer pads without wire bond inductance.
- **TRL Calibration:** De-embedding calibration technique utilizing Thru, Reflect, and Line standards on substrate.
- **Y-Factor Method:** Standard microwave technique for measuring Noise Figure via hot/cold noise ratio.

Frequently Asked / Viva Questions

Q1. What is the function of a Vector Network Analyzer (VNA)?

A VNA measures the complex scattering parameters (S -parameters: magnitude and phase) of microwave components over a swept frequency range.

Q2. Why is calibration required before performing VNA measurements?

Calibration removes systematic errors introduced by test cables, connectors, attenuators, and internal VNA directional couplers.

Q3. Differentiate between SOLT and TRL calibration.

SOLT uses Short, Open, Load, and Thru coaxial standards. TRL uses Thru, Reflect, and Line planar standards and is preferred for MMIC on-wafer probing because it does not require a $50\ \Omega$ matched load standard.

Q4. What is an On-Wafer GSG Probe?

A Ground-Signal-Ground coplanar RF probe that makes direct electrical contact with MMIC pads on the semiconductor wafer up to 110 GHz.

Q5. Explain the Y-Factor method for Noise Figure measurement.

The Y-Factor method measures output noise power when a noise source is switched HOT (T_H) and COLD (T_C). The ratio $Y = N_{HOT}/N_{COLD}$ determines effective noise temperature T_e and Noise Figure NF .

Q6. Define Excess Noise Ratio (ENR) of a noise source.

$ENR = (T_H - T_0)/T_0$, expressed in dB. It quantifies the calibrated noise power output above standard room temperature ($T_0 = 290\text{ K}$).

Frequently Asked / Viva Questions (contd.)**Q7. What is a microwave test fixture?**

A mechanical housing that holds an unencapsulated MIC substrate and provides low-reflection transitions between coaxial cables and planar microstrip lines.

Q8. What is reference plane shift in S-parameter measurements?

The mathematical offset applied by VNA calibration to move the measurement plane from the coaxial cable tips directly to the MIC substrate ports.

Q9. Why is Noise Figure measurement sensitive to receiver gain?

High receiver gain prevents internal meter noise from masking the small noise power differences ($N_{\text{HOT}} - N_{\text{COLD}}$) generated by the Device Under Test.

Q10. What is the role of de-embedding in MIC testing?

De-embedding mathematically subtracts the known parasitic S -parameters of test fixtures and launches from the measured data to isolate the intrinsic MIC device response.